26 A plane circular coil of 30 turns, each of diameter 1.0 cm, rotates 1800 times each minute about a diameter which is perpendicular to a uniform magnetic field of flux density 4.0×10^{-5} T.

What is the maximum instantaneous value of the induced electromotive force?

A 1.2×10^{-6} V **B** 2.8×10^{-6} V **C** 1.8×10^{-5} V **D** 2.3×10^{-5} V

Ans: C

Induced electromotive force, $E = -\frac{d\Phi}{dt} = -\frac{d(NBA\cos\omega t)}{dt} = \omega NBA\sin\omega t$

Maximum induced electromotive force, $E_{max} = \omega NBA = (2\pi f)NBA$

$$= \left(2\pi\frac{1800}{60}\right)(30)\left(4.0\times10^{-5}\right)\left(\frac{\pi\left(1.0\times10^{-2}\right)^2}{4}\right)$$

$$= 1.8\times10^{-5}\,\text{V}$$